

EECE 571Q

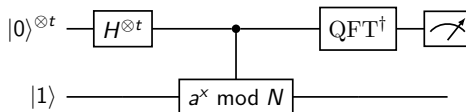
Fermionic Quantum Computation: What, Why, and How to Protect It

Sam Wong

Lecture 1 of 2 — Fermionic Quantum Error Correction (fQEC)

The usual story: algorithms on qubits

One big motivation for quantum computing: solve problems believed to have no efficient classical algorithm — *quantum advantage*. Shor (factoring), Grover (search), ...



Shor: phase estimation reveals the period of $a^x \bmod N$.

The natural unit is the *qubit*, and the target is a *mathematical* problem — the kind an algorithm solves.

But how about simulating quantum systems themselves?

P. W. Shor, SIAM Journal on Computing **26**, 1484 (1997)

Simulating matter means simulating fermions

“Nature isn’t classical... and if you want to make a simulation of Nature, you’d better make it quantum mechanical.”

— Feynman, 1982

Many of the systems we would like to simulate involve *fermions*:

- quantum chemistry — electrons in molecules
- materials — Hubbard models, superconductivity
- high-energy physics — lattice gauge theory

Quantum simulation is one of the most natural things to ask of a quantum computer — and many of its target systems are made of fermions.

R. P. Feynman, International Journal of Theoretical Physics **21**, 467 (1982)

Bosons, fermions — and why a qubit is neither

Identical particles are indistinguishable, so exchanging two can only multiply the state by a phase — and exchanging twice must do nothing. In 3D that forces the phase to be ± 1 :

Bosons (+1)

Integer spin, symmetric. Any number share a mode.

Photons, phonons.

Swap +: $\frac{1}{\sqrt{2}}(|ab\rangle + |ba\rangle)$.

Fermions (−1)

Half-integer spin, antisymmetric. At most one per mode (Pauli). Electrons.

Swap −: $\frac{1}{\sqrt{2}}(|ab\rangle - |ba\rangle)$;
 $a=b \Rightarrow 0$.

Bosons, fermions — and why a qubit is neither

Identical particles are indistinguishable, so exchanging two can only multiply the state by a phase — and exchanging twice must do nothing. In 3D that forces the phase to be ± 1 :

Bosons (+1)

Integer spin, symmetric. Any number share a mode.

Photons, phonons.

Swap $+$: $\frac{1}{\sqrt{2}}(|ab\rangle + |ba\rangle)$.

Fermions (−1)

Half-integer spin, antisymmetric. At most one per mode (Pauli). Electrons.

Swap $-$: $\frac{1}{\sqrt{2}}(|ab\rangle - |ba\rangle)$;
 $a=b \Rightarrow 0$.



Bosons, fermions — and why a qubit is neither

Identical particles are indistinguishable, so exchanging two can only multiply the state by a phase — and exchanging twice must do nothing. In 3D that forces the phase to be ± 1 :

Bosons (+1)

Integer spin, symmetric. Any number share a mode.

Photons, phonons.

Swap $+$: $\frac{1}{\sqrt{2}}(|ab\rangle + |ba\rangle)$.

Fermions (−1)

Half-integer spin, antisymmetric. At most one per mode (Pauli). Electrons.

Swap $-$: $\frac{1}{\sqrt{2}}(|ab\rangle - |ba\rangle)$;
 $a=b \Rightarrow 0$.

A *qubit* is a distinguishable two-level system at a fixed address — nothing happens when you exchange two of them.

A fermionic mode is empty or occupied: two states, like a qubit.
 The exchange sign is what makes it not one.

Roadmap

- ❶ **Local fermionic modes & Jordan–Wigner**
the model, and the precise sense in which qubits are not fermions
- ❷ **Majorana fermions**
half a mode, and a universal gate set
- ❸ **Errors & first codes**
Majorana stabilizer codes, phase and loss errors, and the bridge to Lecture 2

First: which *kind* of fermionic QEC are we doing?

Two perspectives on fermionic QEC

Fermions for qubits

Logical *qubits* from fermionic hardware: keep a mode's two halves far apart, and the hardware itself suppresses errors.

Motive: *protection*.

local operator on one half:



Native fermionic QEC

Logical *fermions* from physical fermions.

Motive: *simulation* — fermionic problems without the mapping tax.

Their mark: logicals *anticommute* across codeblocks, $\{c_a, c_b^\dagger\} = \delta_{ab}$.



This module: the native approach.

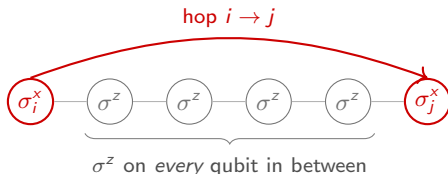
A. Schuckert et al., (2025), arXiv:2411.08955 [quant-ph], Fig. 1a

The mapping tax

Simulating fermions on qubits: one hopping term becomes a Pauli *string*,

$$a_i^\dagger a_j + a_j^\dagger a_i \mapsto \frac{1}{2} \left(\sigma_i^x \sigma_{i+1}^z \cdots \sigma_{j-1}^z \sigma_j^x + \sigma_i^y \sigma_{i+1}^z \cdots \sigma_{j-1}^z \sigma_j^y \right)$$

weight $O(m)$ on m modes — and depth and gate count inflate with it, compounding against fault tolerance.

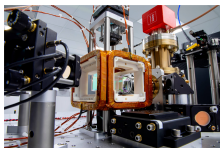


P. Jordan and E. Wigner, Zeitschrift für Physik **47**, 631 (1928)

A. Schuckert et al., (2025), arXiv:2411.08955 [quant-ph], opening (overhead framing)

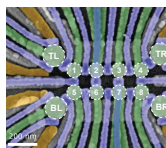
The escape hatch: native fermions

Let the hardware be made of fermions:



neutral atoms in tweezers

QuEra Computing



quantum dots

Hsiao *et al.*, PRX **14**, 011048 (2024)

Antisymmetry is supplied by the hardware, not simulated — no string to pay for.

Question of the lecture:
can we compute — and error-correct — natively?

A. Schuckert *et al.*, (2025), arXiv:2411.08955 [quant-ph], abstract

Segment 2: what *is* a fermionic computer?

Where we are. We want to compute with fermions directly — so we need to say precisely what that means.

What is the model, and where exactly do qubits fail?

- the local fermionic mode model, and what its algebra does
- the precise sense in which qubits are *not* fermions
- a universal gate set — and the one gate that will matter most

The local fermionic mode (LFM) model

m sites, each empty or occupied by a spinless fermion:

$$|n_0, n_1, \dots, n_{m-1}\rangle, \quad n_j \in \{0, 1\}$$

spans the 2^m -dimensional Fock space.

Everything is built from $a_j, a_j^\dagger$, subject to the canonical anticommutation relations:

$$\{a_j, a_k^\dagger\} = \delta_{jk} \mathbb{1}, \quad \{a_j, a_k\} = \{a_j^\dagger, a_k^\dagger\} = 0$$

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 2

The relations, at work

Two modes — what do the anticommutators actually *do*?

Same mode twice. $\{a_0^\dagger, a_0^\dagger\} = 0 \Rightarrow (a_0^\dagger)^2 = 0$ — Pauli exclusion is a *consequence* of the algebra, not a separate rule.

Two modes, two orders.

$$a_0^\dagger a_1^\dagger = -a_1^\dagger a_0^\dagger$$

Filling in the other order costs a minus sign — the exchange sign from the opening, now in operator form.

But occupations commute.

$$[n_0, n_1] = 0, \quad n_j = a_j^\dagger a_j$$

Generators remember the order; observables don't.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 2

Why observables are even

You cannot make a fermion out of nothing — fermions come and go in *pairs*. Why?

Turn the lab through 2π : nothing physically happened. But each fermion contributes a -1 :

$$\begin{array}{ccc}
 & \text{rotate the lab by } 2\pi & \\
 |\text{even}\rangle & \xrightarrow{\quad\quad\quad} & +|\text{even}\rangle \\
 |\text{odd}\rangle & \xrightarrow{\quad\quad\quad} & -|\text{odd}\rangle
 \end{array}$$

So the do-nothing operation *is* parity, and no observable can detect it:

$$R(2\pi) = (-1)^{\sum_j n_j} = P, \quad [\mathcal{O}, P] = 0 \implies \mathcal{O} \text{ is even}$$

Odd operators are not observables — rotational invariance forces it.

G. C. Wick et al., Physical Review **88**, 101 (1952)

What superselection means

A superselection rule splits the state space into *sectors* that no observable connects:

$$\langle \text{even} | \mathcal{O} | \text{odd} \rangle = 0 \quad \text{for every observable } \mathcal{O}$$



So the relative phase across sectors is *unmeasurable* — such a “superposition” is indistinguishable from a classical mixture.

Even — physical: $n_j, a_i^\dagger a_j, a_i a_j$ **Odd — not:** $a_j, a_j^\dagger$

Not “hard to prepare” — *meaningless*: that superposition and that mixture are the same physical state.

G. C. Wick et al., Physical Review **88**, 101 (1952)

S. B. Bravyi and A. Y. Kitaev, Annals of Physics **298**, 210 (2002), Sec. 2

Locality: the precise statement

- The *generators* a_j, a_k anticommute at arbitrary distance — the structural difference from Pauli generators.
- But *physical* operators are the even-parity ones (parity superselection) — and those commute at a distance, just like qubit observables.

The algebras of physical observables look alike; the generating sets don't. Simulation cost is paid at the generator level.

Why the tensor-product identification fails

As *Hilbert spaces*, Fock space $\cong m$ qubits — identify

$$|n_0, n_1, \dots, n_{m-1}\rangle \equiv |n_0\rangle \otimes |n_1\rangle \otimes \dots \otimes |n_{m-1}\rangle$$

so the failure is *not* dimensional.

Under this identification, a one-mode fermionic operator does *not* become a one-qubit operator: the sign $(-1)^{n_0 + \dots + n_{k-1}}$ in a_k touches every site below k .

The spaces match; the local operator structures don't.

The Jordan–Wigner transformation

Task. Represent a_k on qubits, reproducing

$$a_k |n_0, \dots, n_{k-1}, 1, \dots\rangle = (-1)^{n_0 + \dots + n_{k-1}} |n_0, \dots, n_{k-1}, 0, \dots\rangle$$

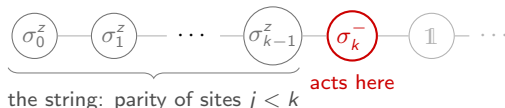
Result.

$$a_k \mapsto \left(\prod_{j < k} \sigma_j^z \right) \sigma_k^-$$

The string records the parity of every site *strictly below* k .

In the occupation basis $|n\rangle$, $n \in \{0, 1\}$:

$$\sigma^+ = |1\rangle\langle 0| \text{ (creates),} \quad \sigma^- = |0\rangle\langle 1| \text{ (annihilates),} \quad \sigma^z |n\rangle = (-1)^n |n\rangle$$

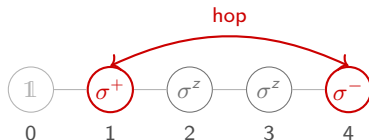


P. Jordan and E. Wigner, Zeitschrift für Physik **47**, 631 (1928)

Worked example: five modes

On $m = 5$ modes:

$$a_1^\dagger a_4 \mapsto \sigma_1^+ \sigma_2^z \sigma_3^z \sigma_4^-$$



Physical reading: the string measures the *parity of the occupation you hop over* — the amplitude's sign depends on how many fermions sit between sites 1 and 4.

The *Pauli weight* — the number of non-identity tensor factors — is 4 here; a hop across k sites costs weight $\sim k$.

Where the string comes from

On $m = 5$ modes, take the hop $a_1^\dagger a_4$. Give each factor its own string:

$$a_1^\dagger = \sigma_0^z \sigma_1^+, \quad a_4 = \sigma_0^z \sigma_1^z \sigma_2^z \sigma_3^z \sigma_4^-$$

Different sites commute, so collect site by site:

$$a_1^\dagger a_4 = \underbrace{\sigma_0^z \sigma_0^z}_{= \mathbb{1}} \underbrace{\sigma_1^+ \sigma_1^z}_{= \sigma_1^+} \sigma_2^z \sigma_3^z \sigma_4^-$$

Site 0 squares away; site 1 absorbs its own σ^z , since $\sigma^+ \sigma^z = \sigma^+$.

The shared string below the *lower* site cancels — only the segment *between* the two sites survives.

A universal gate set for LFMs

BK's four gates, universal for parity-preserving operators (with ancillas), for any modes $i \neq j$:

$$\left\{ \begin{array}{ll} \exp\left(i\frac{\pi}{4} a_i^\dagger a_i\right), & \exp\left(i\frac{\pi}{4} (a_i^\dagger a_j + a_j^\dagger a_i)\right), \\ \exp\left(i\frac{\pi}{4} (a_j a_i + a_i^\dagger a_j^\dagger)\right), & \exp\left(i\pi a_i^\dagger a_i a_j^\dagger a_j\right) \end{array} \right\}$$

number phase · hopping · pairing · interaction

Why these four — each closes a gap:

- **hopping** tunnels a particle between sites, **number phase** is an on-site energy — together, every single-particle rotation
- **pairing** leaves the fixed- N sectors — provably unavoidable
- **interaction** breaks Gaussianity — without it, classically simulable

Universality: their commutators fill the whole even algebra.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), eqs. (13)–(15)

Why four gates suffice

The engine. Gates compose into commutators:

$$e^{tA} e^{tB} e^{-tA} e^{-tB} = e^{t^2[A, B]} + O(t^3)$$

Do A , do B , undo A , undo B : if they commuted you would be back at the start. What is left over *is* the commutator — a direction the generators alone do not span.

Example — hopping routes. For distinct i, j, k :

$$\left[a_i^\dagger a_j, a_j^\dagger a_k \right] = a_i^\dagger a_k$$

neighbouring hops generate a hop between *any* two modes.

Layered. Quadratics (phase, hopping, pairing) close among themselves — all Gaussian, all matchgate. The quartic interaction is what escapes that closure.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), eqs. (13)–(15)

Parity, not particle number

All four gates commute with total parity $P = (-1)^{\sum_j n_j}$ — superselection demands it.

But the pairing gate does *not* conserve particle number $N = \sum_j n_j$: it changes N by ± 2 .

Parity is the conserved quantity of fermionic computation. Particle number is not.

The pair-creation gate

$$\exp\left(i\frac{\pi}{4}(a_j a_i + a_i^\dagger a_j^\dagger)\right)$$

changes particle number by ± 2 — and it is *necessary*: without a non-number-conserving gate, the set is not universal.

Why this matters for QEC. Number-conserving gates never leave a fixed- N sector. But a code whose logical object is a *fermion* cannot live in one sector — its codestates superpose particle numbers (Segment 4).

No number-conserving circuit can prepare a logical *fermion*. This gate is the way in.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), eqs. (14)–(15)

Segment 3: a better language

Where we are. A model, a universality result, and a gate set — but four gates, one of them a *four-body* interaction.

Is there a more natural language for fermions?

Splitting each mode into two Hermitian halves will:

- collapse four gates into two
- expose what is genuinely hard — and what is free
- hand us the measurement that error correction will need

Majorana fermions: half a mode

Split each mode k into two *Hermitian* halves:

$$\gamma_k = a_k + a_k^\dagger, \quad \tilde{\gamma}_k = -i(a_k - a_k^\dagger)$$

All $2m$ halves anticommute alike — enumerate them as $(c_0, c_1, c_2, \dots) \equiv (\gamma_0, \tilde{\gamma}_0, \gamma_1, \dots)$:

$$\{c_j, c_k\} = 2\delta_{jk} \quad \implies \quad c_j^2 = \mathbb{1}$$

Hermitian: $\gamma^\dagger = \gamma$ — a Majorana is its own antiparticle (creating one = annihilating one). Nothing is lost — the map inverts:

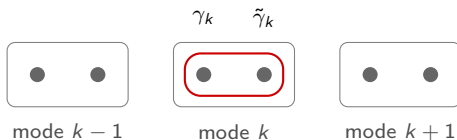
$$a_k = \frac{1}{2}(\gamma_k + i\tilde{\gamma}_k), \quad a_k^\dagger = \frac{1}{2}(\gamma_k - i\tilde{\gamma}_k)$$

m modes $\longleftrightarrow$ $2m$ Majorana halves.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), eqs. (24)–(25)

Occupation is a Majorana bilinear

$$a_k^\dagger a_k = \frac{1}{2} (\mathbb{1} + i \gamma_k \tilde{\gamma}_k)$$



n_k : the same-site pair

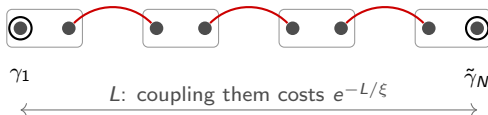
Occupation is a *same-site Majorana pair*. This identity carries the rest of the module.

Why separated Majoranas are protected

Claim. Two Majoranas kept far apart form a decoherence-protected qubit.

Parity superselection: a lone Majorana is odd, hence unphysical — the environment can only couple to Majorana *pairs*, and pair coupling is exponentially small in their separation.

bulk Majoranas bind *across* sites



“Keep two halves of a qubit far apart and the qubit can’t decohere.”

S. B. Bravyi and A. Y. Kitaev, Annals of Physics **298**, 210 (2002), Sec. 6

A. Y. Kitaev, Physics-Uspekhi **44**, 131 (2001)

A universal Majorana gate set

Two gates suffice

$$\left\{ \exp\left(\frac{\pi}{8} c_a c_b\right), \exp\left(i\frac{\pi}{4} c_a c_b c_c c_d\right) \right\}$$

for *any* distinct Majoranas: one bilinear, one quartic.

Three of the four LFM gates are bilinear rotations in disguise:

$$\text{number phase:} \quad n_i = \frac{1}{2}(\mathbb{1} + i\gamma_i \tilde{\gamma}_i)$$

$$\text{hopping:} \quad a_i^\dagger a_j + \text{h.c.} = \frac{i}{2}(\gamma_i \tilde{\gamma}_j - \tilde{\gamma}_i \gamma_j)$$

$$\text{pairing:} \quad a_j a_i + \text{h.c.} = -\frac{i}{2}(\gamma_i \tilde{\gamma}_j + \tilde{\gamma}_i \gamma_j)$$

The same two bilinears — difference or sum. They share no Majorana, so they commute and each exponential *factorises* into bilinear rotations.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), eq. (26)

The real partition: quadratic vs. quartic

Quadratic exponentials only *rotate the modes* — in the flat labels c_j (all $2m$ halves in one list):

$$U c_j U^\dagger = \sum_k \beta_{jk} c_k, \quad \beta \in SO(2m)$$

These are the *matchgate* circuits: classically simulable, hence not universal.

The quartic gate is the unavoidable non-Gaussian ingredient — BK prove no gate on ≤ 3 Majoranas substitutes.

The dividing line of fermionic computation is polynomial degree: quadratic is free; quartic is where the power lives.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), eq. (27), Sec. 6

Trading the quartic gate for a measurement

The quartic *gate* — a four-body interaction, and physically hard — can be traded for a **nondestructive measurement** of

$$M = \gamma_i \tilde{\gamma}_i \gamma_j \tilde{\gamma}_j$$

Why M can be measured at all:

- *even* (four Majoranas) — so it is a physical observable at all
- $M^\dagger = M$, and $M^2 = \mathbb{1}$ — so the outcome is a single bit, ± 1

Nondestructive: it projects onto the ± 1 eigenspace, $P_\pm = \frac{1}{2}(\mathbb{1} \pm M)$ — returning *only* that bit. Whatever superposition lives inside the eigenspace survives untouched.

What the measurement asks

Feed in the workhorse identity, $i\gamma_k\tilde{\gamma}_k = 2n_k - 1$:

$$M = -(2n_i - 1)(2n_j - 1)$$

$$M = -1$$

$$|00\rangle \quad |11\rangle$$

$$n_i + n_j \text{ even}$$

$$M = +1$$

$$|01\rangle \quad |10\rangle$$

$$n_i + n_j \text{ odd}$$

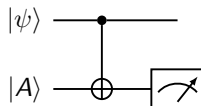
M reports the *parity* of the two modes — and nothing about *which* one is occupied. Each sector stays two-dimensional: a qubit's worth of state survives inside.

It learns a joint property and nothing about the individual state.
That is exactly what a stabilizer does.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 6–7

Reminder: how a T gate gets done

In a fault-tolerant qubit code the Cliffords are easy and T is not. Nobody engineers a T gate — they *inject* it.



- the magic state $|A\rangle = T|+\rangle$ is prepared *offline* — that is where the difficulty is parked
- consume it with one Clifford and one measurement
- outcome 0: done. outcome 1: apply S — also a Clifford

The hard part moves offline into a *resource*. Online, every operation is an easy one.

S. Bravyi and A. Kitaev, Physical Review A **71**, 022316 (2005)

The same trick for Majoranas

	T gate	quartic gate
easy set	Cliffords	bilinear gates & measurements
the resource	magic state $ A\rangle$	one weight-4 measurement
correction	S	bilinear rotations
ancilla ends	consumed	returned intact

The recipe. Ancilla Majorana pair c_4, c_5 in a known state:

- ① measure $c_0 c_1 c_3 c_4$ — the weight-4 resource, with the *ancilla* standing in for c_2
- ② measure $c_2 c_4$ — a bilinear, which releases the ancilla again
- ③ correct, using bilinear rotations
 $\Rightarrow \exp(i\frac{\pi}{4} c_0 c_1 c_2 c_3)$, with the ancilla pair given back.

The hardness sits in the *measurement*, not in a consumable state.

S. B. Bravyi and A. Y. Kitaev, Annals of Physics **298**, 210 (2002), Sec. 7

Segment 4: what goes wrong

Where we are. We can compute natively — and we can measure an even Majorana monomial without disturbing the state.

What are the errors, and can we see them coming?

- two error types — one familiar, one genuinely fermionic
- that same measurement, now doing duty as a *stabilizer*
- a first code — and how it catches an error

Stabilizers are Majorana monomials

A *Majorana monomial* is a product of distinct Majoranas, $c_{j_1} c_{j_2} \cdots c_{j_k}$; its *weight* is k , and even weight $\Rightarrow$ physical. ($\gamma_i \tilde{\gamma}_i$ has weight 2.)

Reframe Segment 3's primitive: a nondestructive measurement of an even Majorana monomial *is* a stabilizer measurement.

- Pick commuting even Majorana monomials $S_1, \dots, S_r$.
- The codespace is their joint $+1$ eigenspace.

Codes whose stabilizers are Majorana monomials: *Majorana stabilizer codes*.



S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 7

Error type I: phase errors

Physical origin: differential light shifts (dephasing) — the *dominant* error channel in neutral atoms, orders of magnitude more frequent than loss.

Algebraic form:

$$E = n_i = \frac{1}{2} (\mathbb{1} + i \gamma_i \tilde{\gamma}_i)$$

the *same-site Majorana bilinear* — Segment 3's workhorse identity, now wearing an error hat. Even, hence physical.

Why “phase”? n_i reads 0 on an empty site, 1 on a full one:

$$e^{-i\theta n_i} : \quad \alpha |0\rangle_i + \beta |1\rangle_i \mapsto \alpha |0\rangle_i + \beta e^{-i\theta} |1\rangle_i$$

Populations untouched — only the *relative phase* moves. This is a Z error: $\sigma_i^Z = -i \gamma_i \tilde{\gamma}_i$.

A. Schuckert et al., (2025), arXiv:2411.08955 [quant-ph], “Repetition code for phase errors”

Error type II: atom loss

Physical origin: the atom leaves the trap.

Algebraic form:

$$E = a_i = \frac{1}{2}(\gamma_i + i\tilde{\gamma}_i)$$

Linear in Majoranas — **odd**, where the phase error was an even bilinear. Odd flips fermion parity, so it is not a legal operator on the system alone: the environment must take the fermion. A Pauli error *rotates* the particle; this one *removes* it.

From here we go after **phase** errors — they dominate, and they are what the code on the next slide corrects.

The repetition code — the codespace

On m modes, take the $m - 1$ *cross-site* checks

$$S_i = i \tilde{\gamma}_i \gamma_{i+1}, \quad i = 1, \dots, m - 1$$

No two share a Majorana, so they all commute — and 2^m states cut by $m - 1$ independent checks always leaves a *two-dimensional* codespace.

Solving them for $m = 3$:

$$|0\rangle_L = \frac{1}{2} (|000\rangle - |110\rangle - |101\rangle - |011\rangle) \quad 0 \text{ or } 2 \text{ particles}$$

$$|1\rangle_L = \frac{1}{2} (|100\rangle + |010\rangle + |001\rangle - |111\rangle) \quad 1 \text{ or } 3 \text{ particles}$$

Conventions follow S25: their $\tilde{\gamma}_i = -i(a_i^\dagger - a_i)$, and higher index filled leftmost.

Both codestates superpose particle numbers — forced: fixed- N codestates $\Rightarrow$ no logical fermion (S25 Theorem).

A. Schuckert et al., (2025), arXiv:2411.08955 [quant-ph], eqs. (5)–(6) & Theorem

The repetition code — the logical fermion



Every Majorana finds a partner *except* the two at the ends — those carry the logical mode:

$$c^\dagger = \frac{1}{2}(\gamma_1 + i\tilde{\gamma}_m), \quad c^\dagger |0\rangle_L = |1\rangle_L, \quad c^\dagger |1\rangle_L = 0$$

The logical fermion sits on the two *edge* Majoranas — Segment 3's separated halves, now by design.

A. Schuckert et al., (2025), arXiv:2411.08955 [quant-ph], eqs. (5)–(6)

A. Y. Kitaev, Physics-Uspekhi **44**, 131 (2001)

How the code sees a phase error

A phase error is a *same-site* pair; the checks are *cross-site* pairs:

$$E_i \sim \gamma_i \tilde{\gamma}_i \quad S_i = i \tilde{\gamma}_i \gamma_{i+1}$$

Two Majorana bilinears anticommute exactly when they share *one* Majorana — so an error flips every check that overlaps it:

	S_1	S_2
no error	+	+
E_1 (shares $\tilde{\gamma}_1$)	−	+
E_2 (shares γ_2 and $\tilde{\gamma}_2$)	−	−
E_3 (shares γ_3)	+	−

Three errors, three distinct syndromes — the code does not just *detect* the phase error, it *locates* it.

A. Schuckert et al., (2025), arXiv:2411.08955 [quant-ph], “Repetition code for phase errors”

Where we got to

- A fermionic mode is empty or occupied — two states, like a qubit. The *exchange sign* is what makes it not one.
- Physical operators are *even*. That single rule priced the mapping tax, protected separated Majoranas, and fixed the shape of every code we wrote.
- Quadratic is free, quartic is where the power lives — and the quartic gate can be bought with a *measurement*.
- Two errors: phase, even and familiar; loss, odd and genuinely fermionic. The repetition code sees the first.

Questions?

Backup: the setup

Assumed cheap: bilinear rotations $\exp(\theta c_j c_k)$, bilinear measurements $c_j c_k$, ancilla preparation — and *one* weight-4 measurement. Outcomes may steer later steps.

The ancilla (Majoranas c_4, c_5), BK eq. (28):

$$(c_4 + i c_5) |\Psi\rangle = 0 \quad \Longleftrightarrow \quad c_4 c_5 |\Psi\rangle = i |\Psi\rangle$$

and since $c_4 c_5 = i(\mathbb{1} - 2n)$, that just says *the ancilla mode is empty*.

Rearranged, it is the tool we actually use:

$$c_4 |\Psi\rangle = -i c_5 |\Psi\rangle$$

Not an operator identity — but against *this* state, c_4 may always be traded for c_5 .

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 7, eq. (28)

Backup: which two operators — and why

Measure $A = c_0 c_1 c_3 c_4$ (the weight-4 resource), then $B = c_2 c_4$ (a bilinear).

	c_0	c_1	c_2	c_3	c_4	c_5
A	●	●		●	●	
B			●		●	
BA	●	●	●	●	×	

shared
⇒ cancels

= $-M$

$$BA = c_2 c_4 c_0 c_1 c_3 c_4 = -c_0 c_1 c_2 c_3 = -M$$

$A^2 = \mathbb{1}$, so its outcome is $z = \pm 1$; $B^2 = -\mathbb{1}$, so its outcome is $y = \pm i$. The projectors are $\Pi_z^{(4)} = \frac{1}{2}(\mathbb{1} + zA)$ and $\Pi_y^{(2)} = \frac{1}{2}(\mathbb{1} - yB)$.

The ancilla sits in *both* — it squares away, so the two measurements multiply to the target.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 7

Backup: the ingredients

Write $C = c_2 c_5$. Note $M^2 = \mathbb{1}$, but $C^2 = (c_0 c_1)^2 = -\mathbb{1}$, so:

$$e^{i\frac{\pi}{4}M} = \frac{1}{\sqrt{2}}(\mathbb{1} + iM), \quad e^{\frac{\pi}{4}C} = \frac{1}{\sqrt{2}}(\mathbb{1} + C), \quad e^{\frac{\pi}{2}c_0 c_1} = c_0 c_1$$

Four reductions on $|\Psi\rangle$ — each one application of $c_4 |\Psi\rangle = -i c_5 |\Psi\rangle$:

$$\begin{aligned} C|\Psi\rangle &= iB|\Psi\rangle, & CA|\Psi\rangle &= iM|\Psi\rangle, \\ CB|\Psi\rangle &= i|\Psi\rangle, & CM|\Psi\rangle &= iA|\Psi\rangle \end{aligned}$$

A generator squaring to $-\mathbb{1}$ behaves like i — and at $\pi/2$ it degenerates into a bare monomial.

Backup: the cancellation

Take the outcome $(z, y) = (+1, +i)$, so $\Pi_{+1}^{(4)} = \frac{1}{2}(\mathbb{1} + A)$ and $\Pi_{+i}^{(2)} = \frac{1}{2}(\mathbb{1} - iB)$. Expand them, using $BA = -M$:

$$(\mathbb{1} - iB)(\mathbb{1} + A) = \mathbb{1} + A - iB + iM$$

Now multiply by $(\mathbb{1} + C)$ and reduce every term:

$\mathbb{1} \rightarrow \Psi\rangle$	$A \rightarrow A \Psi\rangle$	$-iB \rightarrow -iB \Psi\rangle$	$iM \rightarrow iM \Psi\rangle$
$C \rightarrow iB \Psi\rangle$	$CA \rightarrow iM \Psi\rangle$	$-iCB \rightarrow \Psi\rangle$	$iCM \rightarrow -A \Psi\rangle$

A and B — the two *measured* operators — cancel; $\mathbb{1}$ and iM each double.

$$\underbrace{(\mathbb{1} + C)}_{\text{you apply this what the measurement did}} \underbrace{(\mathbb{1} - iB)(\mathbb{1} + A)}_{\text{you apply this what the measurement did}} |\Psi\rangle = 2(\mathbb{1} + iM) |\Psi\rangle = 2\sqrt{2} e^{i\frac{\pi}{4}M} |\Psi\rangle$$

The measurement alone leaves $\Pi_{+i}^{(2)} \Pi_{+1}^{(4)} |\Psi\rangle$ — not yet the gate.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 7, eq. (29)

Backup: the four outcomes

The other projectors run identically, but flip the sign of iM :

$$(\mathbb{1} + C)(\mathbb{1} - iB)(\mathbb{1} - A) |\Psi\rangle = 2(\mathbb{1} - iM) |\Psi\rangle \propto e^{-i\frac{\pi}{4}M} |\Psi\rangle$$

outcome (z, y)	1. rotate ancilla	you then hold	2. then apply
(+1, +i)	$e^{+i\frac{\pi}{4}C}$	$e^{+i\frac{\pi}{4}M}$ ✓	—
(-1, +i)	$e^{+i\frac{\pi}{4}C}$	$e^{-i\frac{\pi}{4}M}$	iM
(+1, -i)	$e^{-i\frac{\pi}{4}C}$	$e^{-i\frac{\pi}{4}M}$	iM
(-1, -i)	$e^{-i\frac{\pi}{4}C}$	$e^{+i\frac{\pi}{4}M}$ ✓	—

Step 1 happens on *every* branch — only its sign depends on y . Step 2 is needed only when you hold the inverse, and iM flips it:

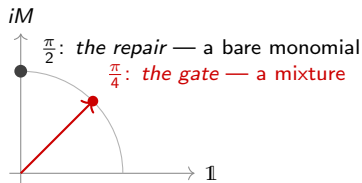
$$iM(\mathbb{1} - iM) = iM + M^2 = \mathbb{1} + iM.$$

After that rotation you hold the gate *or its inverse*, each with probability $\frac{1}{4}$.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 7, eq. (29)

Backup: why the repair is cheaper than the gate

A bad outcome must be repaired with iM . If that were as hard as the gate itself, we would have gained nothing.



Since $e^{i\theta M} = \cos \theta \mathbb{1} + i \sin \theta M$:

$\theta = \frac{\pi}{4}$: $\frac{1}{\sqrt{2}}(\mathbb{1} + iM)$ a mixture — does *not* factorise

$\theta = \frac{\pi}{2}$: $iM = (c_0 c_1)(c_2 c_3)$ two bilinears — free

Fixing a wrong outcome is *easier* than doing the gate.

S. B. Bravyi and A. Y. Kitaev, *Annals of Physics* **298**, 210 (2002), Sec. 7, eq. (29)

Backup: the same structure for T gates

Magic-state injection leaves $T|\psi\rangle$ or $T^\dagger|\psi\rangle$ — so its two branches differ by

$$T/T^\dagger = T^2 = S$$

	qubits	Majoranas
easy set	Clifford	bilinear rotations
the gate	T	$e^{i\frac{\pi}{4}M}$
in the easy set?	<i>no</i>	<i>no</i>
its square	$T^2 = S$	$(c_0c_1)(c_2c_3)$
in the easy set?	yes	yes

Both gates sit *outside* the easy set — but both squares sit inside.
That is what makes a random outcome affordable.